



Spendly

Offline-First Zero-Knowledge Peer-to-Peer Finance Ecosystem

Semester VII · Information Technology · Major Project

[Student Name] · [Enrollment Number] · [College Name]



**CENTRALIZED CLOUD
DATA LEAKAGE**
EXPOSED SURFACE
TRUST DEPENDENCY



**ZERO-KNOWLEDGE
ENCRYPTION**
NO DATA EXPOSURE
TRUSTLESS VERIFICATION

THE PROBLEM

The Financial World Is Broken

Every transaction today leaves a trace — stored on servers you don't control, exposed to breaches you can't prevent, and dependent on infrastructure you can't trust. The current model is fundamentally flawed.

Privacy at Risk

Centralized databases expose sensitive financial data to third parties, advertisers, and malicious actors.

Cloud Dependency

Offline environments render modern payment systems completely useless — a critical failure in emerging markets.

Data Breaches

Centralized storage is a single point of failure. One breach exposes millions of users simultaneously.

POS Costs

Traditional point-of-sale hardware demands monthly subscriptions and proprietary infrastructure.

THE SOLUTION

Introducing the Spendly Ecosystem

Three purpose-built applications working in concert — each designed for a distinct role, all unified by a shared zero-knowledge, offline-first architecture. Together, they redefine how money moves.

Spendly

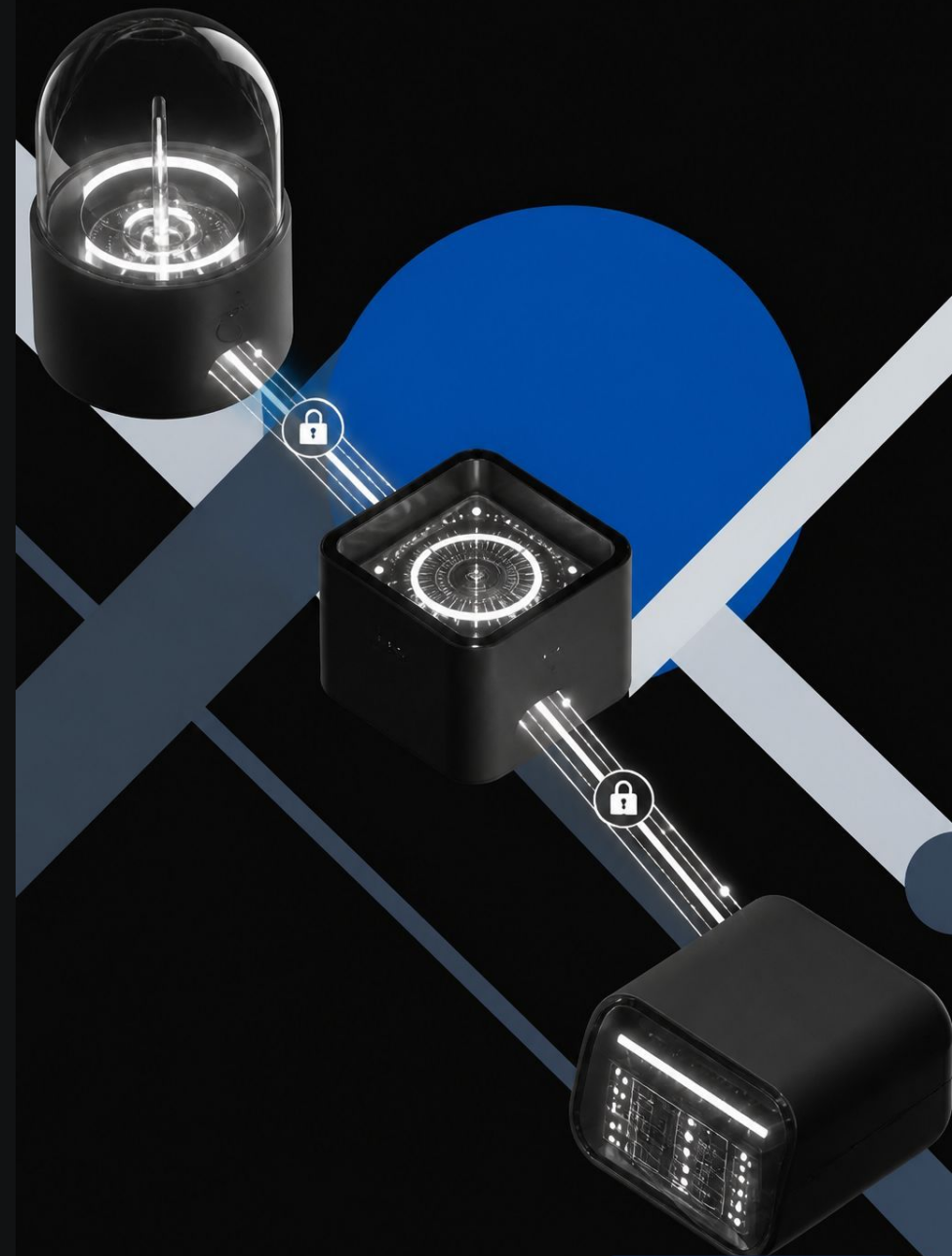
The core personal finance application. Track expenses, manage wallets, generate invoices, and maintain complete financial sovereignty — entirely offline.

Spendly Shop

A merchant-facing point-of-sale interface. Accept payments, manage inventory, and issue QR invoices without any internet connection or monthly fees.

Spendly Radar

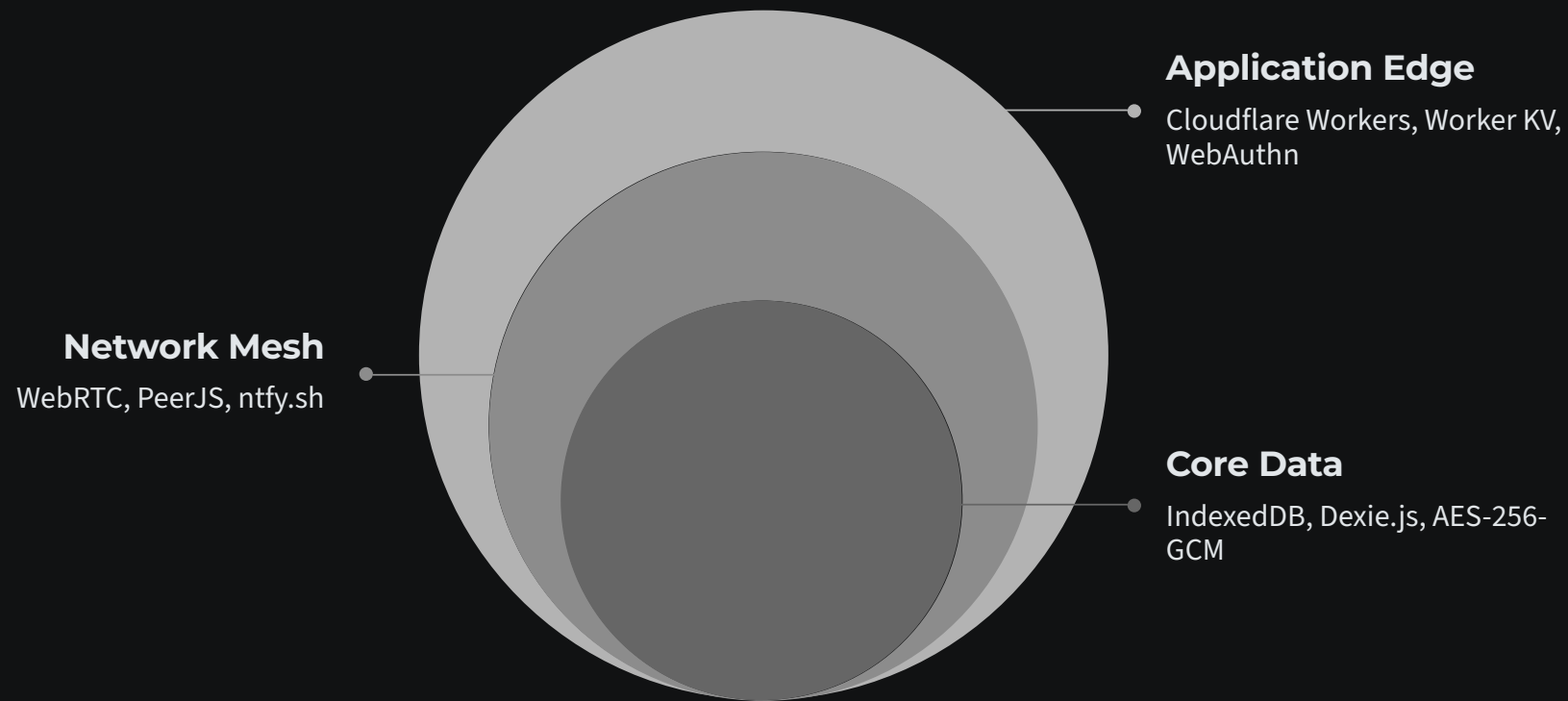
A peer-to-peer discovery and communication layer. Find nearby merchants, claim offers, and synchronize data securely over local networks.



Made with GAMMA

System Architecture

Spendly is engineered as a layered, privacy-preserving stack — each component chosen for its ability to operate without a central server while maintaining cryptographic guarantees.



The architecture flows upward from persistent local storage through encrypted peer-to-peer networking, serverless edge compute, and finally a responsive React-based progressive web application — all operating without a traditional backend.

WORKFLOW

How Spendly Works

From discovery to settlement, every step in the Spendly workflow is designed to be fast, private, and fully offline-capable. No servers. No latency. No compromise.

Discover Offers

Spendly Radar scans for nearby merchant promotions via local peer-to-peer discovery.

Claim Coupon

Offers are claimed and stored locally — encrypted and tied to your identity via WebAuthn.

Scan Product

Merchant scans product barcode or SKU. Inventory lookup happens entirely on-device.

Offer Validation

Applicable coupons are validated against the cart in real time with zero server calls.

Invoice Generation

A signed QR invoice is generated locally and presented to the customer instantly.

Invoice Sharing

Invoice is shared peer-to-peer via WebRTC and stored in IndexedDB for both parties.

Local Storage

All transaction records persist offline in encrypted IndexedDB with Dexie.js indexing.

Analytics

Spending insights and merchant reports are computed locally — no data ever leaves the device.

TECHNOLOGY

Technology Stack

Every technology in the Spendly stack was selected for a single reason: the ability to deliver a premium, secure, and fully offline experience without a traditional server.

Frontend

React · Vite · Tailwind CSS · Zustand ·
Framer Motion

Backend

Cloudflare Workers · Worker KV

Networking

WebRTC · PeerJS · ntfy.sh

Security

WebAuthn · AES-256-GCM

Database

IndexedDB · Dexie.js



SECURITY

Zero-Knowledge by Design

Spendly's security model ensures that no party — not the developer, not the merchant, not the network — ever sees your financial data. Privacy is not a feature. It is the foundation.



Zero-Knowledge Architecture

Transactions are validated cryptographically without revealing any underlying data to any third party.



Offline-First Design

All core functionality operates without internet. Data never traverses an external network unless explicitly shared.



WebAuthn + Biometric Unlock

Device-level biometric authentication via WebAuthn ensures only the legitimate user can access financial data.



AES-256-GCM Encryption

All stored data is encrypted at rest using AES-256-GCM — the same standard used by governments and financial institutions.

FEATURES

Key Features

Spendly consolidates every financial interaction — personal and commercial — into a single, privacy-preserving ecosystem. No subscriptions. No cloud. No compromise.



Expense Tracking

Real-time categorization and visualization of all personal spending — computed entirely on-device.



Merchant Billing

Full point-of-sale functionality with QR invoice generation, no internet required.



Inventory Management

Track stock levels, SKUs, and product metadata locally with instant lookup.



Cash Wallet

Digital cash ledger for offline peer-to-peer transfers and balance tracking.



EMI Reminder

Automated reminders for installment payments, managed without external notifications.



QR Invoice

Cryptographically signed invoices shared via QR code — verifiable without a server.



Offline Synchronization

Seamless peer-to-peer data sync when devices reconnect — no data loss, ever.



Analytics

Privacy-safe spending insights and merchant reports generated locally from encrypted data.

Advantages & Future Scope

Current Advantages

- **Offline-First:** Fully functional without internet connectivity
- **Zero Server Cost:** No infrastructure expenses for merchants or users
- **Privacy:** Zero-knowledge architecture — no data ever leaves your device
- **Fast Billing:** Instant invoice generation with no network latency
- **Secure Communication:** Encrypted peer-to-peer channels via WebRTC

Future Scope

- **AI Analytics:** On-device machine learning for spending pattern insights
- **Voice Billing:** Voice-activated merchant commands for hands-free operation
- **NFC Payments:** Contactless hardware-based peer-to-peer transactions
- **UPI Integration:** Seamless bridge to India's Unified Payments Interface
- **GST Compliance:** Automated tax invoice generation for registered merchants
- **Multi-Store:** Centralized management for merchant chains and franchises

CONCLUSION

The Future of Private Finance

Spendly is more than a project — it is a statement. A declaration that financial privacy, offline resilience, and peer-to-peer trust are not luxuries reserved for the few, but rights owed to everyone. Built without servers. Protected by mathematics. Designed for the world as it is.

Privacy-First

Zero-knowledge architecture ensures your data is yours alone.

Offline-Ready

Works anywhere — no internet, no cloud, no compromise.

Peer-to-Peer

Direct, encrypted communication between users and merchants.

Thank You

Questions & Answers